Multimodal Classification of Ingestate Types Using Wearable Acoustic Sensing

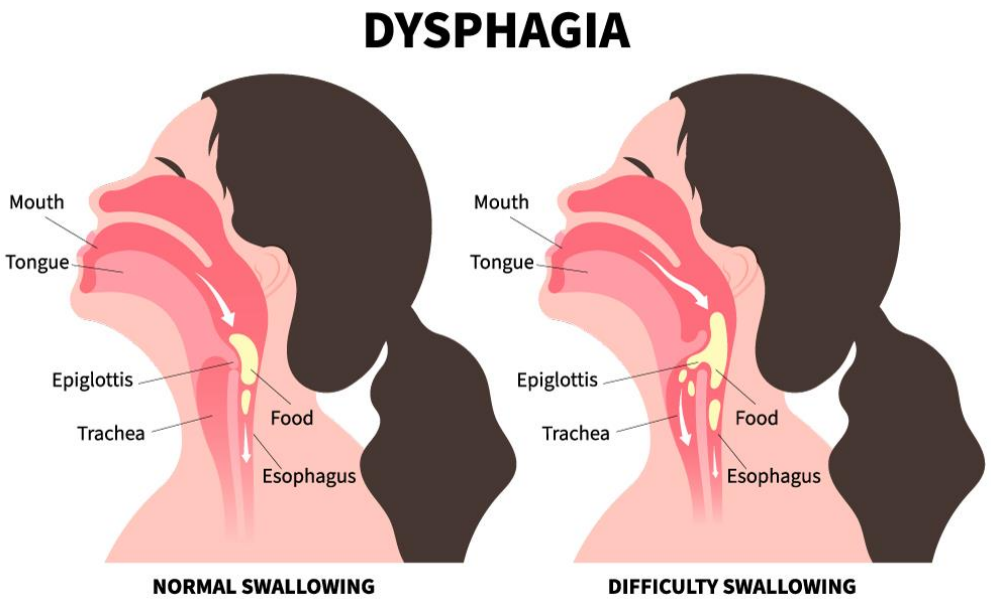
EE499 – Wearable Technologies

April 2026

Daniela Hernández, Dr. Kyle N Winfree

NAU NORTHERN ARIZONA UNIVERSITY

Motivation



Current methods:

- Dysphagia → aspiration, malnutrition
- Clinical methods are invasive and limited
- Need for continuous monitoring
- Wearables enable noninvasive sensing [1], [6]

Current Limitations:

- Wearables detect swallowing events [1]
- Do NOT classify ingestate type
- Food texture strongly affects swallowing [4], [5]

Hypothesis

Different ingestates will produce distinct acoustic patterns, and solids will be easier to classify than liquids.

System Overview

Audio → Preprocessing → Detection → Features → PCA/KNN → Classification

Data

Collection

- .m4a files from Garage Band
- Labels extracted from filename
    - Type: liquid, semi-solid, solid, saliva
    - Size: small, medium, large

Preprocessing

```
%% preprocessing
[b,a] = butter(4, 20/(Fs/2), 'high');
x = filtfilt(b,a,x);

x = x(round(0.1*Fs):end);
```

Steps:

- Convert to mono
- High-pass filter (remove anything under 20Hz)
- Trim first 0.1 seconds

Swallow Detection

Swallow Detection

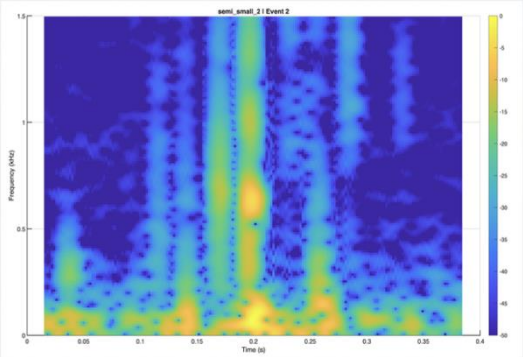
- Short-time energy
- Peak detection
  - Threshold = 0.2
  - Min distance = 0.7 s
- **Over-segmentation observed**

```
%% swallow detection
windowSize = round(0.02*Fs);
energy = movmean(x.^2, windowSize);

if max(energy) == 0
    continue;
end

energy = energy / max(energy);
energy_smooth = movmean(energy, 150);

threshold = 0.2;

[~, locs] = findpeaks(energy_smooth, ...
    'MinPeakHeight', threshold, ...
    'MinPeakDistance', round(0.7*Fs));

fprintf('Detected swallows: %d\n', length(locs));
```

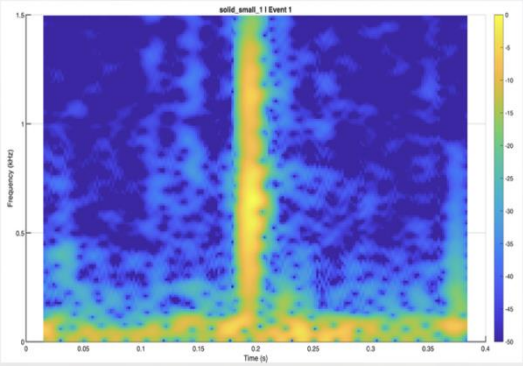
Spectrogram

Analysis

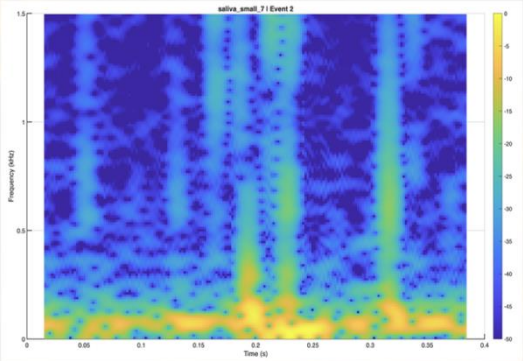
- Time-frequency representation
- Liquids → sharp bursts
- Solids → broader energy spread
- Based on acoustic signal differences [6], [7]



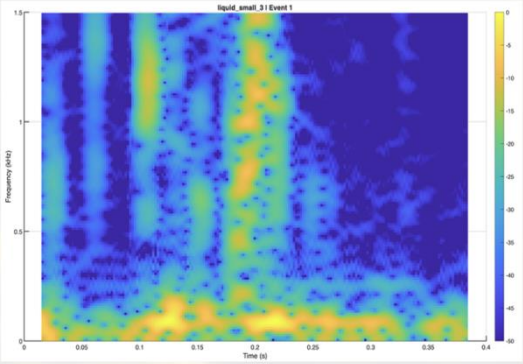
Spectrogram of one small semi-solid event , relatively easily recognized by the program.



Spectrogram of one small solid event , relatively easily recognized by the program.



Spectrogram of one small saliva event, recognized by the program with some error.



Spectrogram of one small liquid event, recognized by the program with some error

Feature Extraction

Time-domain:

- RMS energy
- Zero-crossing rate
- Duration

Frequency-domain:

- Centroid
- Bandwidth
- Entropy
- Flatness

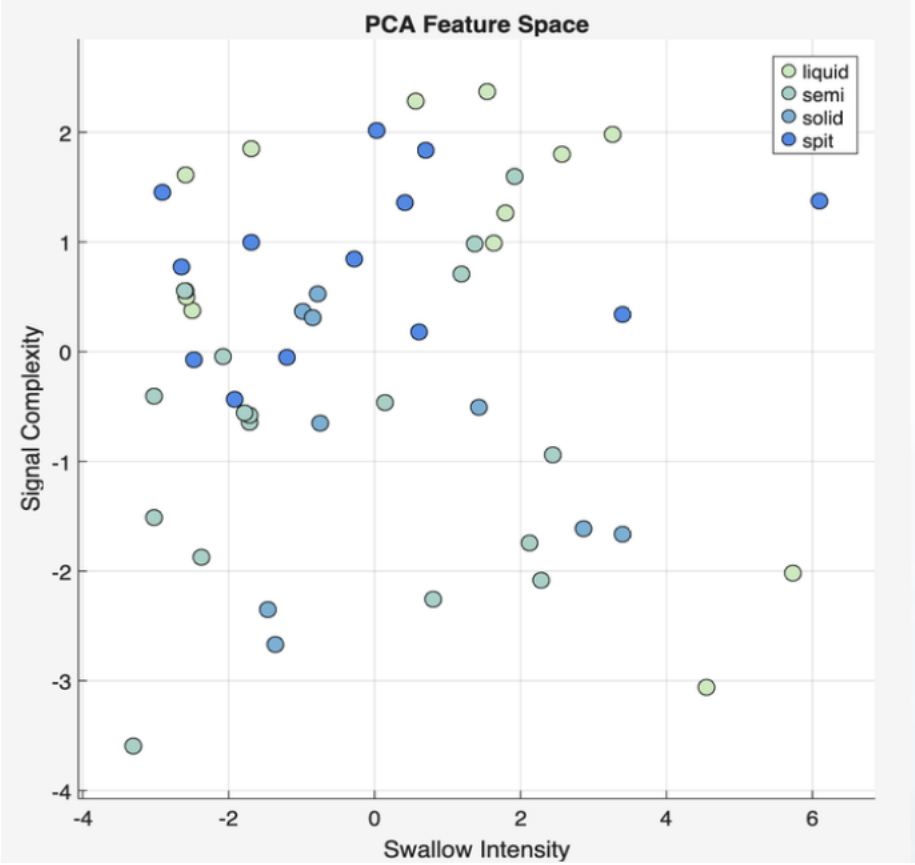
PCA

Dimensionality Reduction

- Reduces feature space
- Retains 95% variance
- Enables visualization

PCA Visualization

- Liquid & saliva → distinct clusters
- Semi & solid → overlap



KNN

K-Nearest Neighbors (KNN)

- k = 7 neighbors
- Majority vote classification

```
%% KNN
k_val = 7;

true_labels = [];
pred_labels = [];

fprintf('\n--- RESULTS ---\n');

for i = 1:size(data,1)

    test = data(i,1:end-1);
    train = data;
    train(i,:) = [];

    dists = sqrt(sum((train(:,1:end-1) - test).^2,2));
    [~, idx] = sort(dists);

    neighbors = train(idx(1:k_val),end);
    pred = mode(neighbors);

    true_labels(end+1) = data(i,end);
    pred_labels(end+1) = pred;

    fprintf('True: %-6s | Pred: %-6s\n', ...
        types{data(i,end)}, types{pred});
end
```

Results

- Accuracy ≈ 65%
- Above chance (25%)

Confusion Matrix

Liquids → high accuracy

Saliva → good accuracy

Semi vs solid → confusion

	Liquid	Saliva	Semi	Solid
Liquid	76.9%	15.4%	0.0%	7.7%
Saliva	18.8%	50.0%	18.8%	12.5%
Semi	11.1%	27.8%	38.9%	22.2%
Solid	0.0%	9.1%	63.6%	27.3%

Key Findings

- Acoustic signals are informative
- Some ingestates are distinguishable
- Overlap limits accuracy

Limitations

- Small dataset
- Over-segmentation
- Feature overlap
- Simple model (KNN)

Future Work

- Larger dataset
- Improved features
- Multimodal sensors
- Advanced models

Conclusion

- Ingestate classification is possible
- Accuracy ≈ 65%
- Strong foundation for wearable monitoring

Swallowing sounds contain meaningful structure, but similar ingestates remain difficult to separate.

Thank you

Citations

[1] D. Hossain, J. G. Thomas, M. A. McCrory, J. Higgins, and E. Sazonov,
"A Systematic Review of Sensor-Based Methods for Measurement of Eating Behavior," *Sensors*, 2025.

[2] H. Kalantarian, N. Alshurafa, T. Le, and M. Sarrafzadeh,
"Monitoring Eating Habits Using a Piezoelectric Sensor-Based Necklace," *Computers in Biology and Medicine*, 2015.

[3] M. K. O'Brien et al.,
"Advanced Machine Learning Tools to Monitor Biomarkers of Dysphagia: A Wearable Sensor Proof-of-Concept Study," *Digital Biomarkers*, 2021.

[4] M. T. Murillo-Arbizu et al.,
"Textural Classification of Commercial Foodstuffs for Dysphagia Using Back-Extrusion Test," *Foods*, 2025.

[5] M. C. Wong et al.,
"Quantitative Textural and Rheological Data on Texture-Modified Foods and Thickened Liquids Classified Using IDDSI Guidelines," *Foods*, 2023.

[6] B. P.-H. So et al.,
"Swallow Detection with Acoustics and Accelerometric-Based Wearable Technology: A Scoping Review," *International Journal of Environmental Research and Public Health*, 2022.

[7] Y. Song et al.,
"Multimodal Deep Ensemble Classification System with Wearable Vibration Sensor for Detecting Throat-Related Events," *npj Digital Medicine*, 2025.